

> PRACTICE EXERCISE

Cell : The Unit of Life

- Which of the following organism does not obey the 'cell theory'?
(a) Virus (b) Bacteria
(c) Fungi (d) Plants
- The nucleic acid in a cell was discovered by
(a) E Strasburger (b) A Kossel
(c) F Miescher (d) Huxley
- The 'cell theory' for organisms was proposed by
(a) Purkinje and Von Mohl
(b) Schleiden and Schwann
(c) Carolus Linnaeus
(d) Felix Dujardin
- The fact that new cells arise from pre-existing cells was proposed by
(a) JE Purkinje (b) F Dujardin
(c) R Virchow (d) E Abbe
- The scientists who first recognised and named the nucleus was
(a) Robert Hooke
(b) Robert Brown
(c) Theodore Schwann
(d) Thomas Morgan
- Which of the following is a physical basis of life?
(a) Mitochondria (b) Ribosome
(c) Protoplasm (d) Nucleus
- The bacteria were discovered by which of the following person?
(a) H Urey
(b) E Haeckel
(c) AV Leeuwenhoek
(d) Ivanowski
- The term 'Meiosis' is coined by
(a) Fleming
(b) Blackmann
(c) Liebig
(d) Farmer and Moore
- The physical basis of life protoplasm was named by
(a) Purkinje (b) J Huxley
(c) Robert brown (d) Sherington
- Which of the following is not present in prokaryotes?
(a) Ribosomes
(b) Nuclear membrane
(c) Cell wall
(d) Plasma membrane

- Which one among the following is known as 'animal starch'?
(a) Cellulose (b) Glycogen
(c) Pectin (d) Chitin
- Consider the given cell organelles.
1. Mitochondria 2. Plastids
3. Ribosome
Which of the cell organelle(s) given above is/are common in both plant and animal cells?
(a) Only 2
(b) 1 and 2
(c) 1 and 3
(d) All of the above
- Rigidity of cell wall is due to
(a) chitin (b) lignin
(c) suberin (d) pectin
- The plasma membrane of animal cells is highly elastic. It is mainly due to the presence of
(a) carbohydrates
(b) proteins
(c) lipids
(d) nucleic acids
- The secretory product pectin of cell wall is chemically a
(a) protein
(b) carbohydrate
(c) lipid
(d) nucleic acid
- Chromosomes are
(a) present only in the nucleus of a cell
(b) the largest in number in human cells
(c) made up of DNA as a main component
(d) visible in all cells at every time
- Consider the following statements
1. Endoplasmic Reticulum (ER) is called as 'skeleton of cell'.
2. ER is divided into three types according to the presence or the absence of ribosome.
3. Smooth ER is rich in actively protein secreting cells.
Which of the statement(s) given above is/are correct?
(a) Only 1
(b) 1 and 2
(c) 2 and 3
(d) Only 3

- Match the following Columns.

Column I (Cell organelle)	Column II (Common names)
A. Ribosome	1. Suicidal bags of cell
B. Lysosome	2. Protein factory of cells
C. Mitochondria	3. Controller of cells
D. Nucleus	4. Power house of cell

Codes

A B C D	A B C D
(a) 2 1 3 4	(b) 2 1 4 3
(c) 4 2 1 3	(d) 1 4 2 3

- Which of the following cell organelles functions as the powerhouse of a living cell?
(a) Chloroplast (b) Mitochondria
(c) Ribosomes (d) Golgi apparatus
- Which one of the following organelles is the smallest membrane bound organelle?
(a) Ribosome (b) Golgi bodies
(c) Lysosome (d) Nucleolus
- When our cells are filled with dead or damaged organelles like mitochondria, ER, etc. The degradation of these organelles is mainly performed by
(a) ribosomes (b) lysosomes
(c) peroxisomes (d) glyoxysomes
- Consider the following statements
1. Cortical granules originate from Golgi bodies.
2. Cortical granules form the fertilisation membrane.
3. Cortical granules are found only in ovum.
Which of the statement(s) given above is/are correct?
(a) Only 2 (b) 1 and 3
(c) 2 and 3 (d) All of these
- Consider the following statements
1. Lysosomes have hydrolytic enzymes.
2. Lysosomes are autophagic.
3. Lysosomes can dissolve proteins, lipids and carbohydrates.
4. All lysosomes are same structurally and functionally.

Which of the statement(s) given above is/are correct?

- (a) Only 1 (b) 1 and 2
(c) 1, 2 and 3 (d) All of these

24. Consider the following statements

1. Chloroplast is known as 'cell within a cell'.
2. Chloroplast is the largest cell organelle found in plant cells.
3. Chloroplasts have single-stranded extra chromosomal DNA.

Which of the statement(s) given above is/are correct?

- (a) Only 2 (b) 1 and 2
(c) 1 and 3 (d) All of these

25. Which of the following pairs of cell organelles mainly concerned with fatty acid metabolism in both plants and animals?

- (a) Ribosomes and nucleus
(b) Lysosomes and mitochondria
(c) Peroxisomes and mitochondria
(d) Glyoxysomes and peroxisomes

26. DNA is mainly found in the following organelles.

1. Mitochondria
2. Chloroplasts
3. Peroxisomes
4. Nucleus

Which of the statements given above are correct?

- (a) 2 and 4 (b) 1, 2 and 4
(c) 2, 3 and 4 (d) All of these

27. Which of the following cell organelles is found in a cell that is very active in protein synthesis?

- (a) Mitochondria
(b) Chloroplast
(c) Lysosome
(d) Ribosome

28. Consider the following organelles.

1. Chloroplast
2. Mitochondria
3. Peroxisome

Which of the organelle(s) given above is/are concerned with photorespiration?

- (a) Only 3 (b) 1 and 2
(c) 2 and 3 (d) All of these

29. Number of chromosomes in *Drosophila* is

- (a) eight (b) ten
(c) twelve (d) fourteen

30. How many linkage groups are present in man?

- (a) 46 (b) 23
(c) 48 (d) 10

31. Which of the following phases is the longest phase in the meiotic cell division?

- (a) Anaphase-II
(b) Telophase-I
(c) Prophase-I
(d) Metaphase-I

32. Which of the following is the correct sequence of the different phases of cell division?

1. Anaphase 2. Telophase
3. Prophase 4. Metaphase

Codes

- (a) 3, 1, 4, 2 (b) 1, 3, 2, 4
(c) 3, 4, 1, 2 (d) 1, 2, 3, 4

33. Chromosomes are best seen in

- (a) telophase (b) anaphase
(c) metaphase (d) prophase

34. Which type of division occurs in cleavage?

- (a) Amitotic (b) Mitotic
(c) Meiotic (d) Both (b) and (c)

35. In which one of the following types of cell divisions crossing over of chromosomes takes place?

- (a) Mitosis (b) Meiosis
(c) Amitosis (d) Cytokinesis

36. Which is the correct order of phases in prophase-I?

- (a) Leptotene, Diakinesis, Pachytene, Diplotene, Zygotene
(b) Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis
(c) Diakinesis, Pachytene, Diplotene, Zygotene, Leptotene
(d) Diakinesis, Pachytene, Leptotene, Diplotene, Zygotene

37. Which of the following stages is associated with the formation of chiasmata during meiotic cell division?

- (a) Leptotene (b) Zygotene
(c) Pachytene (d) Diplotene

38. Which of the following organisms have peptidoglycan compound as an important constituent in cell wall?

- (a) Bacteria and cyanobacteria
(b) Bacteria and unicellular eukaryote
(c) Archaeobacteria and eukaryote
(d) All members of Monera and Protista

39. Which of the following term is associated with the failure of meiosis-II to occur after meiosis-I?

- (a) Brachymeiosis
(b) Dinomitosis
(c) Diminution
(d) Karyokinesis

> Previous Years' Questions

40. Which organelle in the cell, other than nucleus, contain DNA?

☞ 2013 (I)

- (a) Endoplasmic reticulum
(b) Golgi apparatus
(c) Lysosome
(d) Mitochondria

41. Which of the following parts are found in both plant and animal cells?

☞ 2014 (II)

- (a) Cell membrane, chloroplast, vacuole
(b) Cell wall, nucleus, vacuole
(c) Cell membrane, cytoplasm, nucleus
(d) Cell wall, chloroplast, cytoplasm

42. Which one among the following statement is correct?

☞ 2015 (I)

- (a) Prokaryotic cells possess nucleus
(b) Cell membrane is present in both plant and animal cells
(c) Mitochondria and chloroplasts are not found in eukaryotic cells
(d) Ribosomes are present in eukaryotic cells only

43. The living content of cell is called protoplasm. It is composed of

☞ 2016 (I)

- (a) only cytoplasm
(b) cytoplasm and nucleoplasm
(c) only nucleoplasm
(d) cytoplasm, nucleoplasm and other organelles

Classification of Plants and Animals

44. Study of principle and procedures of classification is called

- (a) taxonomy (b) systematics
(c) nomenclature (d) identification

45. Which of the following does not contain chlorophyll?

- (a) Fungi (b) Algae
(c) Bryophyta (d) Pteridophyta

46. Bryophytes are photosynthetic but do not have vascular tissue and true roots. This feature enables them to resemble with which of the following?

- (a) Fungi (b) Algae
(c) Pteridophytes (d) Angiosperms

47. Vascular tissue is found in

- (a) Thallophyta (b) Bryophyta
(c) Pteridophyta (d) Lichens

48. The gymnosperms are

- (a) seed bearing plants
(b) non-flowering
(c) thallus bearing
(d) Both (a) and (b)

49. When any plane passing through the central axis of the body divides the organism into two halves that are approximately mirror images, it is called
 (a) radial symmetry
 (b) bilateral symmetry
 (c) Both (a) and (b)
 (d) None of the above

50. If we sprinkle common salt on an earthworm, it dies due to
 (a) osmotic shock
 (b) respiratory failure
 (c) toxic effect of salt
 (d) closure of pores of skin

51. Itching due to insect bite is caused by
 (a) formic acid
 (b) acetic acid
 (c) lactic acid
 (d) maleic acid

52. Which of the following is the group with closely related animal types?
 (a) Sea star, sea lilly, sea hare
 (b) Dogfish, silverfish, crayfish
 (c) Leech, louse, snail
 (d) Jellyfish, sea fan, sea anemone

53. Which one of the following animals breathe through the skin?
 (a) Fish (b) Pigeon
 (c) Frog (d) Cockroach

54. Match the following Columns.

Column I	Column II
A. Himalayan salamander	1. Amphibia
B. Indian shark	2. Chondrichthyes
C. Sea-horse	3. Osteichthyes
D. Tortoise	4. Reptilia

Codes

A B C D	A B C D
(a) 1 2 3 4	(b) 1 2 4 3
(c) 2 1 4 3	(d) 2 1 3 4

55. Consider the following statements
 1. Heart is three-chambered in fishes.
 2. Heart is four-chambered in birds.
 3. Most animals of class—Amphibia are characterised by two pair of limbs.
 4. In all reptiles respiration is by lungs only.

Which of the above statements are correct?

- (a) 1, 2, 3, 4 (b) 1 and 3
 (c) 2 and 4 (d) 2, 3, 4

56. Match the following Columns.

Column I	Column II
A. Gill slits	1. Birds
B. Scales	2. Osteichthyes
C. Operculum	3. Chondrichthyes
D. Hairs	4. Reptilia
E. Lungs	5. Mammalia

Codes

A B C D E	A B C D E
(a) 3 4 2 5 1	(b) 1 2 4 3 5
(c) 5 4 2 1 3	(d) 1 2 5 4 3

57. Consider the following statements

1. Warm-blooded animals can remain active in cold environment in which cold-blooded animals can hardly move.
 2. Cold-blooded animals require much less energy to survive than warm-blooded animals.

Which of the statement(s) given above is/are correct?

- (a) Only 1 (b) Only 2
 (c) Both 1 and 2 (d) Neither 1 nor 2

58. Bats can fly in dark because they
 (a) have strong wings
 (b) have sharp eyes
 (c) produce ultrasonic waves
 (d) are nocturnal

59. Which one of the following characteristics is common among parrot, *Platypus* and kangaroo?
 (a) Oviparity
 (b) Toothless jaws
 (c) Homeothermy
 (d) Functional postanal tail

➤ Previous Years' Questions

60. Which one among the following groups of animals maintains constant body temperature in changing environmental conditions? **2013 (I)**

- (a) Birds (b) Amphibians
 (c) Fishes (d) Reptiles

61. Among the following animals, choose the one having three pairs of legs. **2014 (I)**

- (a) Spider (b) Scorpion
 (c) Bug (d) Mite

62. Which one among the following is the generic name of the causal organism of elephantiasis? **2014 (II)**

- (a) Filaria
 (b) Microfilaria
 (c) *Wuchereria bancrofti*
 (d) *Culex pipiens*

63. Which one among the following pairs is not correctly matched?

2014 (II)

- (a) Sandal wood — Partial root plant
 (b) *Cuscuta* — Parasite
 (c) *Nepenthes* — Carnivorous
 (d) Mushrooms — Autotroph

64. Match the following Columns.

2014 (II)

Column-I	Column-II
A. <i>Acsaris</i>	1. Mammalia
B. Malarial parasite	2. Arthropoda
C. Housefly	3. Nematoda
D. Cow	4. Protozoa

Codes

A B C D
(a) 3 4 2 1
(b) 3 2 4 1
(c) 1 2 4 3
(d) 1 4 2 3

Genetics, Molecular Biology and Evolution of Life

65. The scientist who is considered as the father of genetics?

- (a) Lamarck
 (b) Charles Darwin
 (c) Hugo de Vries
 (d) Gregor Johann Mendel

66. Dr. Hargobind Khorana is credited for the discovery of
 (a) synthesis of proteins
 (b) synthesis of genes
 (c) synthesis of nitrogenous bases
 (d) None of the above

67. A pair of contrasting characters is called

- (a) phenotype (b) genotype
 (c) allele (d) None of these

68. Which one of the following ratio represents a test cross?

- (a) 3 : 1 (b) 9 : 7
 (c) 1 : 1 : 1 (d) 1 : 1 : 1 : 1

69. Addition of one chromosome to a diploid set is

- (a) trisomy (b) monosomy
 (c) nullisomy (d) tetrasomy

70. A couple with blood types A and B may have children with blood group

- (a) A and B
 (b) A, B and AB
 (c) A, B, AB and O
 (d) Only AB

71. Which one among the following is not correct about Down's syndrome?

- (a) It is a genetic disorder
- (b) Affected individual has early ageing
- (c) Affected person has mental retardation
- (d) Affected person has furrowed tongue with open mouth

72. What is the most conspicuous salient feature of people with 'Progeria'?

- (a) More hairs on body
- (b) Less immunity to opportunistic infections
- (c) Faster rate of ageing
- (d) Suffer from infertility

73. The nucleic acid in cell was discovered by

- (a) E Strasburger
- (b) A Kossel
- (c) F Miescher
- (d) Huxley

74. The sugar in RNA is

- (a) ribose
- (b) deoxyribose
- (c) sucrose
- (d) lactose

75. Which base replaces thymine in RNA?

- (a) Adenine
- (b) Guanine
- (c) Uracil
- (d) Cytosine

76. Transfer of genetic information from one generation to the other is accomplished by

- (a) DNA
- (b) messenger RNA
- (c) transfer RNA
- (d) Both (b) and (c)

77. Chromosomes are

- (a) present only in the nucleus of a cell
- (b) the largest in number in human cells
- (c) made up of DNA as a main component
- (d) visible in all cells at every time

78. The heredity unit gene is made up of

- (a) RNA
- (b) DNA
- (c) protein
- (d) endoplasmic reticulum

79. The correct model of DNA structure is proposed by

- (a) Jacob and Monod
- (b) Watson and Crick
- (c) Khorana
- (d) Baltimore and Temin

80. Enzyme responsible for DNA replication is

- (a) polymerase
- (b) proteolytic
- (c) dehydrogenase
- (d) carboxylase

81. DNA structure was discovered by

- (a) Weismann
- (b) Watson and Crick
- (c) Sutton
- (d) Jacob

82. Which of the following is represented by X in figure?



- (a) Adenine
- (b) Guanine
- (c) Uracil
- (d) Thymine

83. Genetic code is

- (a) singlet
- (b) doublet
- (c) triplet
- (d) None of these

84. Which of the following were first formed?

- (a) Genes
- (b) Cells
- (c) Eobionts
- (d) Coacervates

85. Life originated in

- (a) air
- (b) land
- (c) water
- (d) All of these

86. Genetic variation arises by

- (a) recombination
- (b) mutation
- (c) chromosomal aberrations
- (d) All of the above

87. The evolution of human species took place mainly in

- (a) Asia
- (b) Africa
- (c) Europe
- (d) China

88. The organs which are morphologically different but perform the same function are called

- (a) homologous organs
- (b) analogous organs
- (c) vestigial organs
- (d) None of the above

89. Vermiform appendix is vestigial in man due to

- (a) cellulose digestion
- (b) omnivorous diet
- (c) heterodont condition
- (d) cooking habit

90. Which of the following have not undergone much of a change during the process of evolution over millions of years?

- 1. Crocodile
- 2. Cockroach
- 3. Horse

Codes

- (a) 1 and 2
- (b) 2 and 3
- (c) 1 and 3
- (d) All of these

91. Which of the following is the correct sequence in which the given animal groups appeared on the earth during the course of evolution?

- (a) Porifera — Annelida — Coelenterata — Protozoa
- (b) Protozoa — Coelenterata — Porifera — Annelida
- (c) Annelida — Porifera — Protozoa — Coelenterata
- (d) Protozoa — Porifera — Coelenterata — Annelida

92. A haemophilic man marries with a normal homozygous woman.

What is the probability that their son will be haemophilic?

- (a) 100%
- (b) 75%
- (c) 50%
- (d) 0%

93. The nearest relative of man are

- (a) lemurs
- (b) apes
- (c) new world monkeys
- (d) old world monkeys

94. Which is the correct sequence in order of chronology?

- 1. Rediscovery of Mendel's law of inheritance.
- 2. Darwin's theory of evolution.
- 3. Synthetic theory of evolution
- 4. Hugo de Vries theory of mutation.

Codes

- (a) 1, 2, 3, 4
- (b) 2, 4, 1, 3
- (c) 3, 2, 1, 4
- (d) 4, 2, 3, 1

95. Which one of the following is an example of vestigial organ in man?

- (a) Jaw apparatus
- (b) Ear muscles
- (c) Canine teeth
- (d) Humerus

96. Arrange the following in the order of their evolution.

- 1. Amphibians
- 2. Fish
- 3. Reptiles
- 4. Birds

Codes

- (a) 1, 2, 4, 3
- (b) 2, 1, 4, 3
- (c) 2, 1, 3, 4
- (d) 4, 1, 3, 2

97. Which one of the following theories was not proposed by Charles Darwin?

- (a) Natural selection
- (b) Survival of the fittest
- (c) Struggle for existence
- (d) Inheritance of acquired characters

98. The theory of 'survival of the fittest' was propounded by

- (a) Lamarck
- (b) Mendel
- (c) Charles Darwin
- (d) Karl Landsteiner

99. According to Darwin, variations are

- (a) heritable
- (b) non-heritable
- (c) mutational
- (d) environmental

100. Who proposed the mutation theory of evolution?

- (a) Huxley
- (b) Darwin
- (c) Lamarck
- (d) Hugo de Vries

101. Synthetic theory was proposed by

- (a) Morgan
- (b) Muller
- (c) Darwin
- (d) Dobzhansky

- 102.** Microevolution can be measured by comparing observed allelic frequencies. These can be predicted by
- chance
 - the Hardy-Weinberg equilibrium
 - differential mortality
 - Mendelian ratio

➤ Previous Years' Questions

- 103.** The fossil of *Archaeopteryx* represents the evidence of origin of **☑ 2013 (I)**
- birds from reptiles
 - mammals from reptiles
 - reptiles from amphibians
 - mammals from birds
- 104.** Two strands of DNA are held together by **☑ 2014 (I)**
- hydrogen bonds
 - covalent bonds
 - electrostatic force
 - van der Waals' forces

- 105.** Match Column I with Column II and select the correct answer using the codes given below **☑ 2016 (I)**

Column I (Geological Time-Scale)	Column II (Life form)
A. Pleistocene	1. Mammals
B. Palaeocene	2. Human genus
C. Permian	3. Invertebrates
D. Cambrian	4. Frogs

Codes

A B C D	A B C D
(a) 2 1 4 3	(b) 2 4 1 3
(c) 3 4 1 2	(d) 3 1 4 2

- 106.** Which of the following cause(s) variation in the genetic material of progeny? **☑ 2016 (I)**
- Sexual reproduction
 - Asexual reproduction
 - Mutations
 - Epigenetic changes

Select the correct answer using the codes given below:

- Only 2
- 1, 2 and 3
- 1, 3 and 4
- 1 and 3

The Plant Morphology Physiology

- 107.** Root hairs are the extension of
- epiblema cells
 - cortex cells
 - pericycle cells
 - xylem tracheids

- 108.** Match the following Columns.

Column I	Column II
A. Primary root	1. It is formed due to the repeated division of primary root.
B. Tap root system	2. It is primary root and its branches.
C. Fibrous root	3. Root arises any place other than the root system.
D. Adventitious root	4. It is direct prolongation of the radicle noticed nearly in all dicotyledonous plants.

Codes

A B C D	A B C D
(a) 1 2 3 4	(b) 3 4 2 1
(c) 2 1 4 3	(d) 4 2 1 3

- 109.** Stem develops from
- plumule
 - radicle
 - pericarp
 - procambium
- 110.** Which one of the following statements regarding potato is correct?
- It is a root
 - It is a normal stem
 - It is a modified stem
 - It is a modified root
- 111.** A leaf without petiole is known as
- subpetiolate
 - sessile
 - zygomorphic
 - heteromorous
- 112.** The largest flower in the world is
- lotus
 - Rafflesia*
 - giant cactus
 - None of these
- 113.** The fruit of coconut is
- drupe
 - hesperidium
 - composite fruits
 - berry
- 114.** Which of the following is correctly matched?
- Tomato—Pome
 - Banana—Berry
 - Mango—Berry
 - Apple—Drupe

- 115.** Match the following Columns.

Column I (Plants)	Column II (Seed dispersal mechanism)
A. Coconut	1. By animals
B. Drumstick	2. Explosive mechanism
C. Cocklebur (<i>Xanthium</i>)	3. By water
D. Castor	4. By wind

Codes

A B C D	A B C D
(a) 2 1 4 3	(b) 2 4 1 3
(c) 3 4 1 2	(d) 3 1 4 2

- 116.** In groundnut, the root is
- nodulated
 - napiform
 - epiphytic
 - photosynthetic

- 117.** Which one of the following is dicot endospermic seed?

- Maize
- Bean
- Castor
- Wheat

- 118.** The apple fruit is called as false fruit because

- its pericarp is inconspicuous
- its actual fruit is located within edible fleshy thalamus
- its develops from the inferior ovary
- its endocarp is cartilaginous

- 119.** Wood is the common name for

- vascular bundle
- cambium
- secondary xylem
- secondary phloem

- 120.** The sharp and pointed outgrowths present on the stem of rose are called

- prickles
- thorns
- spines
- hooks

- 121.** Cork cambium is

- primary meristem
- ground meristem
- lateral meristem
- intercalary meristem

- 122.** The characteristic odour of garlic is due to which one of the following?

- Chlorine-containing compounds
- Fluorine-containing compounds
- Nitrogen-containing compounds
- Sulphur-containing compounds

- 123.** Which one of the following nutrients is not a structural component of the plant?

- Nitrogen
- Calcium
- Phosphorus
- Potassium

- 124.** Palak leaves are rich in

- vitamin-A
- vitamin-D
- iron
- carotene

- 125.** Pulse crops can fix atmospheric nitrogen because of

- root nodules
- deep roots
- aerial roots
- root hair

- 126.** The plant which is parasite in nature is

- Cuscuta*
- Utricularia*
- Rhizopus*
- Green plants

- 127.** Out of the following elements which is required in largest quantity?

- Phosphorus
- Nitrogen
- Calcium
- Sulphur

- 128.** Interveneal necrosis in lemon leaf is caused by deficiency of

- boron
- molybdenum
- copper
- zinc